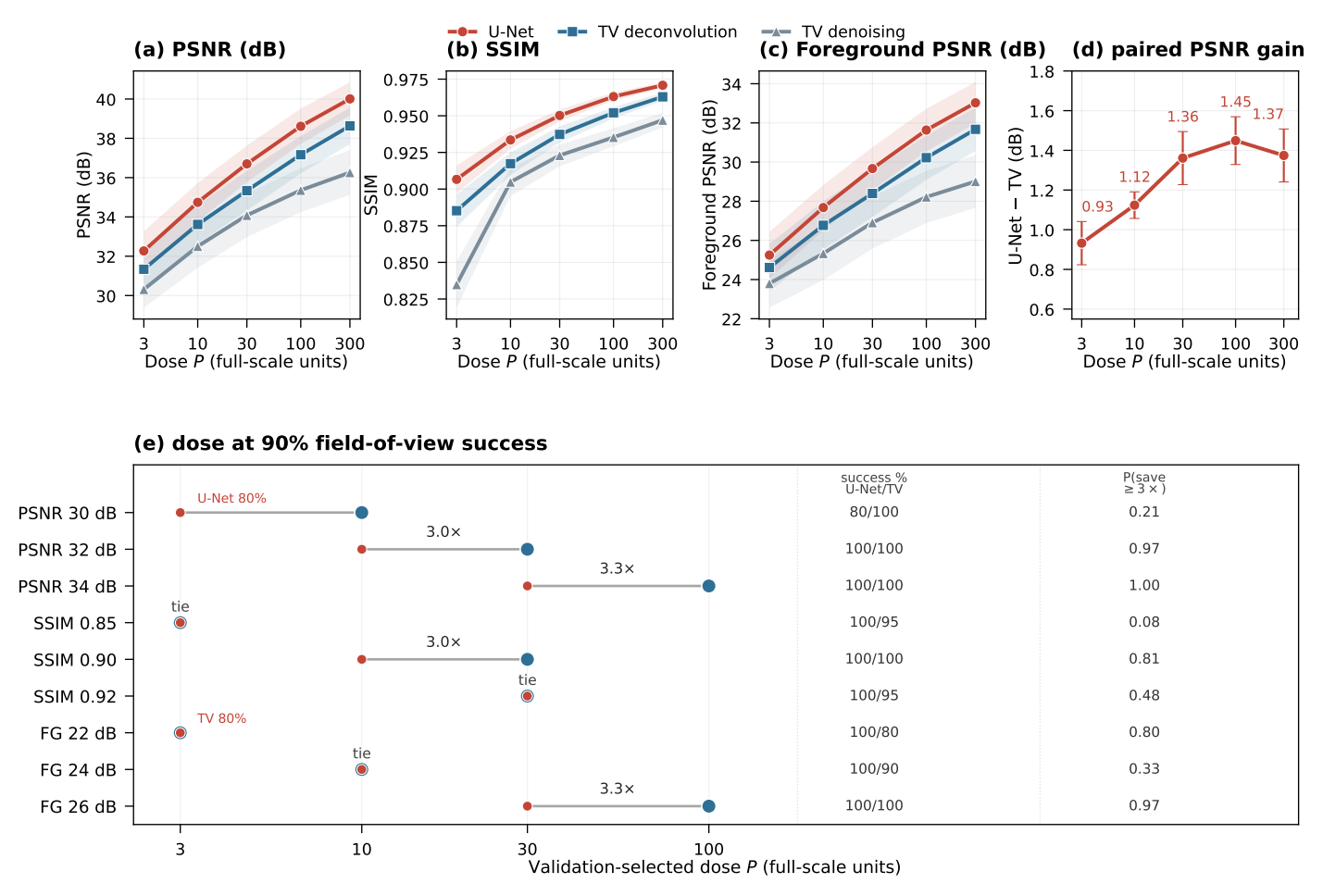
U-Net
TV deconvolution
TV denoising
(a) PSNR (dB)
PSNR (dB)
Dose P (full-scale units)
(b) SSIM
SSIM
Dose P (full-scale units)
(c) Foreground PSNR (dB)
Foreground PSNR (dB)
Dose P (full-scale units)
(d) paired PSNR gain
U-Net − TV (dB)
0.93
1.12
1.36
1.45
1.37
Dose P (full-scale units)
(e) dose at 90% field-of-view success
success % U-Net/TV
P(save ≥ 3×)
PSNR 30 dB
U-Net 80%
80/100
0.21
PSNR 32 dB
3.0×
100/100
0.97
PSNR 34 dB
3.3×
100/100
1.00
SSIM 0.85
tie
100/95
0.08
SSIM 0.90
3.0×
100/100
0.81
SSIM 0.92
tie
100/95
0.48
FG 22 dB
TV 80%
100/80
0.80
FG 24 dB
tie
100/90
0.33
FG 26 dB
3.3×
100/100
0.97
Validation-selected dose P (full-scale units)